Yashkaran Chauhan

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Education

Georgia Institute of Technology — December 2026
Master of Science in Computer Science — *Atlanta, GA*
Concentration: *Artificial Intelligence* — *GPA: 4.00*

Georgia Institute of Technology — December 2025
Bachelor of Science in Computer Science; Faculty Honors — *Atlanta, GA*
Relevant Coursework: *Data Structures & Algorithms, Machine Learning, Deep Learning, NLP* — *GPA: 3.96*

Technical Skills

Languages: Java, Python, C++, MySQL, Javascript, Typescript, SQL, MATLAB, HTML/CSS
Frameworks: React, Angular, Node.js, Express.js, Spring Boot, Spring Security
Libraries: PyTorch, pandas, NumPy, Matplotlib, Sklearn, Sci-py, Transformers, Diffusers, OpenCV, Keras, Tensorflow
Tools & Cloud: AWS (Bedrock, S3), Docker, Jenkins, Git, Slurm, Ollama

Experience

Graduate AI Engineering Intern — May 2026 - August 2026
FINRA — *Rockville, MD*

- Improved the reliability of LLM answers over dense SEC filings, raised measured groundedness to 95% across 1,200 internal chats, by building an automated evaluation pipeline on AWS Bedrock and S3 that scored precision, recall, and LLM-as-a-judge at chunk and document level to guide retrieval and prompt improvements.
- Strengthened executive confidence in the chatbot's ROI by quantifying an average 8 hrs/wk in productivity gains across 100 users, developing an MCP extension that models task-completion time from live conversation context to estimate cumulative time saved.

ML Security Researcher — August 2025 - Present
Georgia Tech — *Atlanta, GA*

- Designing an ML attack in PyTorch against a 105-bit iris-based key derivation scheme, exploiting subsample Hamming distance correlations to reconstruct authenticating feature vectors and test whether the scheme's effective entropy falls below its claimed bound, using a Stable Diffusion model evaluated on the IITD and ND iris datasets.
- Built a modular LangChain + Ollama ML framework supporting end-to-end data ingestion, model execution, evaluation, and inference, improving experimental reliability by 3×, enabling consistent reproduction of attacks and defenses across 20+ configurations in local and cloud-style environments.

Product Engineering Intern — May 2025 - August 2025
VeriSign — *Reston, VA*

- Architected and implemented a Spring Boot RESTful API to automate domain name data retrieval, enabling engineering teams to fine-tune DNS ML systems and generate performance analysis reports for latency checks.
- Integrated Spring Security with a custom login filter and token-based authentication, combined with server-side caching and dynamic rate-limiting to accelerate data transfer from multi-hour manual process to sub-5 second automated responses.

Projects

LLM Security Benchmarking Platform | *Python, LangChain* — January 2026 – April 2026

- Developed a solo agentic benchmarking platform (Python, LangChain, Slurm on GT HPC) implementing 5 attacks and 3 defenses across 5 papers, revealing that 2 attacks defeated all defenses while 1 defense failed against basic prompt injection, grounding ongoing research into dynamic ML security defenses.

Fine-tuning on Adversarial Images | *Python, PyTorch* — March 2025 – May 2025

- Accomplished PyTorch-based adversarial fine-tuning of ResNet50 to recover accuracy on perturbed images by 25% through a GPU-accelerated adversarial-robustness pipeline in Python/PyTorch—streaming ImageNet-1k, performing 20-step PGD attacks on 5K+ samples.

Leadership & Community Engagement

Organizer — May 2026 – August 2026
FINRA Intern Volunteering — *Rockville, MD*

Co-Founder & Instructor — June 2019 – June 2023
Junior Java Summer Camp — *Brambleton, VA*

- Educated 150 students over the summer—achieving 80% retention and 75+ students continuing into CS—by designing a 3-level Python/Java curriculum (JavaFX, PyTorch) with 30+ tailored lessons on data types, OOP, and advanced concepts.